Henry J. Rohde

Address: 255 Iowa Avenue, Iowa City, IA 52240
Mobile: (608) 347-9904 / Email: hjrohde99@gmail.com

Education

University of Iowa, Iowa City, IA Expected May 2029

- Ph.D., Applied Mathematical and Computational Sciences (AMCS)
- Current GPA: 3.4/4.0

University of Wisconsin, Whitewater (UW – W), Whitewater, WI May 2024

- Bachelor of Science, Physics, Graduate School Preparation Emphasis
- Bachelor of Science, Mathematics, Pure Mathematics Emphasis
- Cumulative GPA: 4.0/4.0

Research Experience

Research Assistant, *Stimulated Two-Photon Emission in Astrophysics: A Possible Source of Anomalous Microwave Emission (AME)* Fall 2023 – May 2024

Dr. Robert Benjamin, Department Chair and Professor of Physics, UW – W

- Investigating two-photon emission and applying the phenomenon to interstellar gas clouds
- Deriving the quantum mechanics behind the phenomenon, using Python to compare work to real data

REU Student, *The Fabrication of hBN - Insulated Josephson Junctions* May – July 2023

Dr. Hugh Churchill, Associate Professor of Physics, University of Arkansas – Fayetteville

- Worked with preparing two-dimensional (2D), materials for Josephson junctions
- Performed photolithography to prepare the 2D materials
- Used atomic force microscope to examine the 2D materials
- Trained summer high school interns to continue the project

Mathematics Research Assistant, ***Quiver Representations and Lie Theory*** Fall 2022, Spring 2023

Dr. Xueqing Chen, Professor of Mathematics, UW – Whitewater

- Researched quiver representations, Lie algebras, and cluster algebras to explore the connections between them using linear and abstract algebra

Iceland Travel Study Spring 2022

Dr. Bhattacharyya, Professor of Geology, UW – Whitewater and Dr. Yavuzcetin, Professor of Physics, UW – Whitewater

- Developed an equation for measuring the pressure at the bottom of a geyser
- Worked with a colleague to use an Arduino and pressure sensor to verify that the equation works under real conditions

- Applied equation to the Strokkur geyser at Hveragerði Geothermal Field
- Presented work at the end of travel study to fellow travel study students

Presentations

University of Wisconsin – Whitewater Undergraduate Research Day March 2024

- Poster Presentation: *The Hydrogen Recombination Spectrum and Two – Photon Emission*

University of Arkansas REU May – July 2023

- Oral Presentation: *The Fabrication of hBN-Insulated Josephson Junctions*

National Conference on Undergraduate Research at UW – Eau Claire April 2023

- Oral Presentation: *Quiver Representations and Lie Theory*

University of Wisconsin – Whitewater Undergraduate Research Day March 2023

- Oral Presentation: *Quiver Representations and Lie Theory*

Independent Study

Relativistic Quantum Mechanics Fall 2023

Dr. Abdelkrim Boukahil, Professor of Physics, UW – W

- Studied relativistic quantum mechanics – in particular, studying the Klein – Gordon and Dirac Equations

Work Experience

Teaching Assistant, Mathematics Department, University of Iowa Fall 2024 - present

- Spring 2026 – Math 1020: Elementary Functions
  - Teaching algebra, precalculus, and trigonometry
- Fall 2025 – Math 1005: College Algebra
  - Taught high school algebra
- Spring 2025, Fall 2024 – Math 1350: Quantitative Reasoning for Business
  - Taught algebra, simple and compound interest, annuities, differential calculus

Tutor, Physics Department, UW – W Fall 2022, 2023

- Tutored students for *Methods of Theoretical Physics*
  - Explained the mathematics in detail to students
  - Showed how the concepts can be used in physics, through homework problems

Tutor, Physics Department, UW – W Summer 2021, 2022

- Tutored students for introductory algebra-based physics courses:
  - Taught fundamental physics and algebra concepts
  - Instructed students how to apply the concepts to homework problems

Awards

- *Dean's List – College of Letters and Sciences* Fall 2020 – Spring 2024
- *Birch and Aspen Award* Fall 2023
- *Chancellor Scholar Scholarship* Fall 2020 – Spring 2024
- *Y. W. Fok Scholarship* Fall 2023, Spring 2024
- *2022 Fall Undergraduate Research Grant* Spring 2023
- *2022 Physics Excellence Award in Memory of Spencer Twining* Spring 2022
- *2021 First Year Physics Award* Spring 2021
- *Warhawk Exceptional Academics Award* Fall 2020, Spring 2021

Professional Memberships

- Society for Industrial and Applied Mathematics (SIAM) Fall 2023 – present
- Society of Physics Students (SPS) Fall 2019 – Spring 2022

Skills

- Computer Languages and Software: Python, Julia, LaTeX, MATLAB, and R